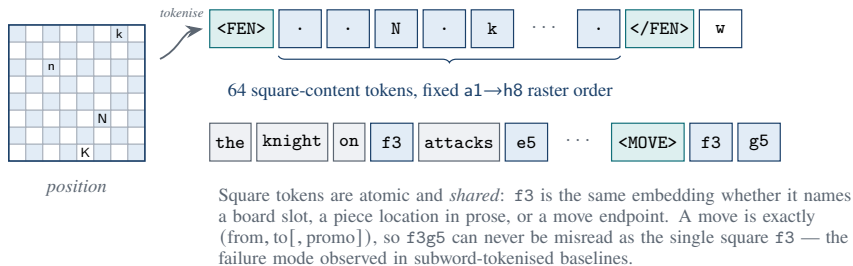
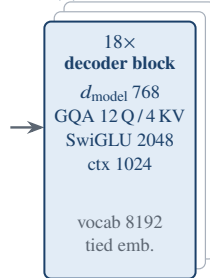


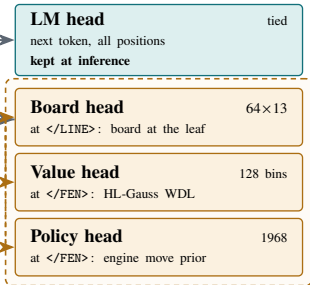
(a) Input encoding



(b) Backbone



(c) Heads



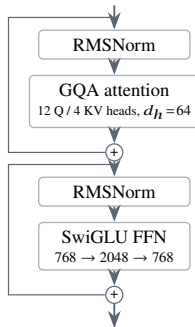
training only — removed at inference.

(d) Dual-scheme rotary encoding



A board token at square s rotates all three bands, carrying its own (rank, file) phase; a prose token zeroes the rank/file bands and reduces to standard 1-D RoPE. File, rank and diagonal neighbours therefore sit at *fixed* rotary offsets (+1 file, +1 rank, +1+1 diagonal), so board geometry is a learnable stride pattern instead of something attention must recover from a run-length-encoded FEN string. The 64-slot raster is emitted in full — no run-length compression — so square i is always at a computable offset from `<FEN>`.

(e) Decoder block



(f) Objective and budget

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_b \mathcal{L}_{\text{board}} + \lambda_v \mathcal{L}_{\text{value}} + \lambda_p \mathcal{L}_{\text{policy}}$$

$\lambda_{\{b,v,p\}}$ are annealed to 0 across the second half of training, so the model is never able to lean on heads it will not have at inference.

119.6M backbone parameters (≈ 120 MB int8);
 2.25 M auxiliary-head parameters, discarded. KV
 cache at full context: 18.9 MB fp16. Training bud-
 get ≈ 3.1 B tokens, $\approx 2.2 \times 10^{18}$ FLOPs, ≈ 40 h on one
 T4.